

Education

[REDACTED]
B.Sc. Computer Science

Aug 2024 – May 2028
[REDACTED]

- Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming (Java), Digital Electronics, Python, Calculus and Vectors, Linear Algebra, Discrete Math, UI/UX, VBA Automation, Database Principles

Experience

Programming and Education Facilitator

Jun 2025 – Sep 2025
[REDACTED]

- Delivered daily instruction to 40+ students per session by teaching programming fundamentals using Edison robots, micro:bit (MakeCode/JavaScript), and Python, improving engagement and concept retention.
- Strengthened classroom effectiveness by designing structured lesson plans, themed activities, and assessment rubrics to track participation and skill mastery.
- Supported 10–15 students concurrently during labs and challenges by providing one-on-one debugging assistance and concept clarification.
- Maintained a safe, positive learning environment while collaborating with instructors and staff, earning consistent positive feedback from parents and program coordinators.

Projects

Plante: Smart Greenhouse and AI Assistant Next.js, TypeScript, Python, Raspberry Pi [GitHub] [Devpost]

- Won Best Designed Award at uOttawaHack 8 by building a gamified pixel art smart greenhouse with integrated hardware control and a web dashboard within a 36-hour hackathon.
- Enabled near real time plant monitoring by streaming temperature, humidity, soil moisture, and light data from a Raspberry Pi through a Flask REST API into a Next.js application.
- Achieved remote physical interaction by wiring SG90 servos through an Arduino using USB serial communication, allowing users to open and close the greenhouse lid from the UI.
- Improved usability of sensor data by integrating a Gemini powered AI assistant with ElevenLabs voice synthesis, weekly summaries, and SMS alerts via Twilio.

Pokémon TCG OpenCV Card Scanner Python, OpenCV, NumPy [GitHub]

- Automated trading card extraction by developing an OpenCV pipeline that detects, isolates, and perspective corrects cards from raw images, processing over 50 cards without manual cropping.
- Built a consistent dataset by standardizing images using adaptive thresholding, background segmentation, and color normalization.
- Optimized OCR performance by reducing single-region processing latency by approximately 36%, lowering inference time from about 2.5 seconds to 1.6 seconds through model selection and pipeline pruning.
- Planned a scalable UI to browse, sort, and filter scanned cards by set, rarity, and attributes to support future collector tools and ML dataset inspection workflows.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, C/C++, SQL, HTML/CSS, VBA, Bash

Frameworks and Libraries: Next.js, React, Node.js, Flask, FastAPI, Tailwind CSS, Framer Motion, Express

AI and Data: Google Gemini API, OpenCV, NumPy, OCR (Tesseract, EasyOCR), Sentiment Analysis

IoT and Systems: Raspberry Pi, Arduino, GPIO, Serial Communication, Sensor Integration

Developer Tools: Git, Docker, AWS, Azure, Sentry, Postman, Twilio API, MongoDB, Mongoose

Concepts: Full Stack Architecture, IoT Systems, API Design, Real Time Data, Agile